
Reproducing Delphi on MIMIC-IV and Extending It with Discrete Phenotype Tokens for Mortality Prediction

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Abstract

We reproduce the Delphi generative EHR transformer of Shmatko *et al.* (2025) on MIMIC-IV and extend it with a discrete text-to-token fusion mechanism. The work has two parts. First, we process MIMIC-IV into a Delphi-compatible longitudinal event format, characterise its sparsity and temporal structure through exploratory data analysis (EDA), train Delphi from scratch on the processed sequences, and reproduce four diagnostic figures from the original paper (per-chapter AUC, embedding UMAP, attention heatmap, and disease-specific risk curves). Second, we present Delphi-Px, a discrete alternative to a failed continuous note-fusion experiment, that extracts 33 clinical phenotype tokens from discharge summaries and inserts them directly into the event timeline. On the MIMIC-IV test split, our reproduction (Delphi-Base) achieves a mean chapter AUC of 0.858 and a Death AUC of 0.680, comparable to the UK Biobank-trained Delphi. Delphi-Px raises Death AUC to 0.721 (+0.041 absolute) while incurring only a -0.007 change in mean chapter AUC. SHAP analysis shows that all 33 phenotype tokens carry positive mean attribution for Death, with palliative-care, DNR, ICU, intubation, sepsis, and AKI tokens supplying clinically interpretable risk evidence.

1 Introduction

The original Delphi paper (Shmatko *et al.*, 2025) demonstrates that a small GPT-2-style transformer trained on structured event sequences can predict the natural history of human disease across UK Biobank cohorts. We address two questions left open by that work for clinical deployment in a US ICU setting: (i) *does the architecture transfer to MIMIC-IV with its sparser longitudinal coverage?* and (ii) *can the structured-only event sequence be augmented with information from discharge summaries in a way that improves mortality prediction without losing interpretability?* We answer the first question by reproducing the Delphi training pipeline on MIMIC-IV and replicating the diagnostic figures of the original paper, and the second question by introducing a discrete phenotype-token extension that we call Delphi-Px.

Contributions. (i) An end-to-end MIMIC-IV preprocessing and EDA pipeline that converts heterogeneous tables into Delphi-compatible event sequences and quantifies temporal sparsity; (ii) a faithful reproduction of the Delphi backbone (5.83 M parameters) on MIMIC-IV with mean chapter AUC of 0.858 and reproductions of the original paper’s figures 2f, 3b, 3c, 3d; (iii) Delphi-Px, a discrete text-to-sequence fusion that adds 33 clinical phenotype tokens and raises Death AUC by +0.041 over the structured-only baseline while preserving auditability.

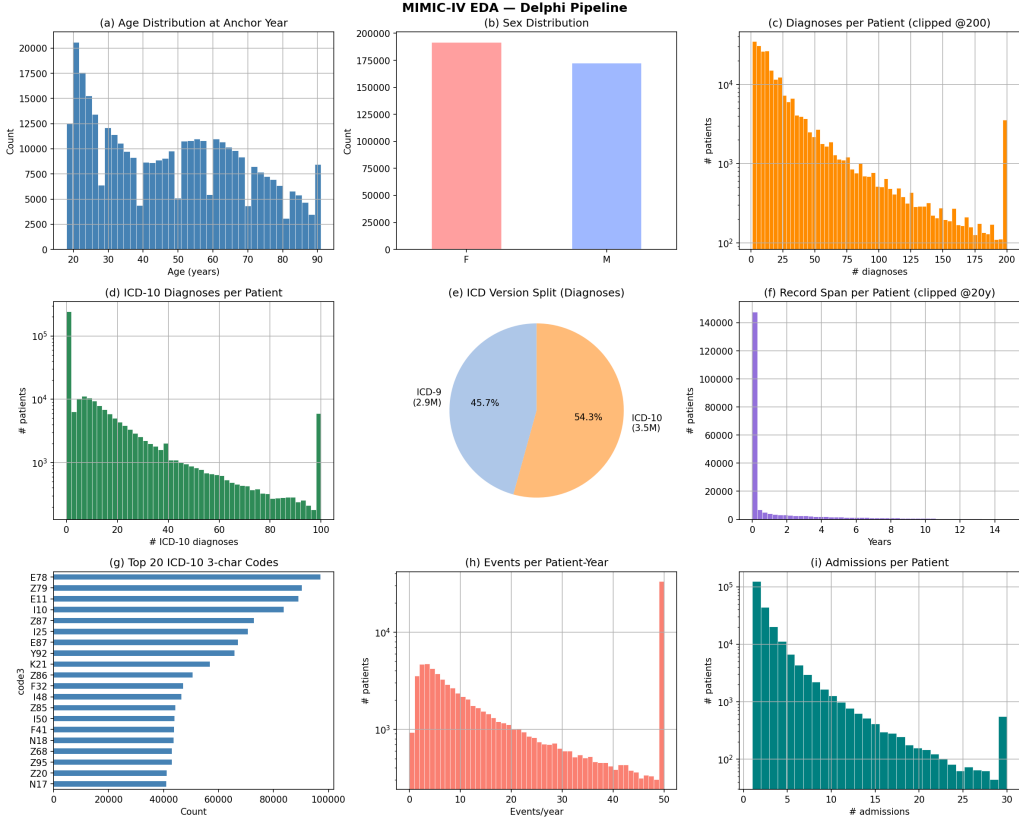


Figure 1: EDA overview on MIMIC-IV: event-count distributions, ICD-version split, and the ICD-10 vocabulary frequency curve. Most patients carry few events; coverage is highly skewed.

2 Exploratory Data Analysis

2.1 Longitudinal sequence construction

We process the MIMIC-IV hosp module (admissions, patients, diagnoses, procedures, laboratory events, transfers) into a Delphi-compatible event sequence. For each patient we order events by absolute age in days, encode each event as a discrete token (ICD-10 diagnoses, binned laboratory observations, demographics, and admission/transfer/discharge sentinels), and GEM-map pre-2015 ICD-9 events to ICD-10 to recover legacy admissions. The resulting cohort contains 364,627 patients of whom 122,905 (33.7%) carry ≥ 1 ICD-10 diagnosis and 38,301 (10.5%) have a recorded death. Final per-split counts are 69,407 / 19,830 / 9,915 patients (train / val / test) yielding 2.59 M, 0.74 M, and 0.20 M structured events.

2.2 Event frequencies and sparsity

The event-frequency distribution is heavy-tailed (Figure ??). Patients hold a median of 13 diagnoses (mean 28.5, p95 102, max 2,396), a median of 1 admission (mean 2.4, p95 7), and a median record span of 0 years – meaning that more than half of patients appear in the data only on one calendar day. Among 6.36M diagnosis events, 54.3% are ICD-10 (post-2015) and 45.7% are GEM-mapped ICD-9 (98.3% mappable, the residual 1.7% being deprecated codes). The ICD-10 vocabulary covers 1,513 codes at ≥ 5 occurrences, 1,085 at ≥ 50 , and 923 at ≥ 100 ; the long tail of rare codes is the principal source of vocabulary noise during fine-tuning. The most frequent ICD-10 chapter heads – E78 (97k), Z79 (90k), E11 (89k), I10 (84k) – match the chronic-disease phenotype expected of an ICU referral population.

2.3 Sequence lengths and temporal gaps

The 95th percentile sequence length is 102 events; the maximum (2,396) is dominated by high-utilisation patients with ≥ 50 admissions. The training block size is therefore set to 128 with overflow events compressed by the SummaryEncoder. The median 0-year record span identifies a population-level temporal sparsity issue: many patients are seen for a single admission with no longitudinal follow-up. This motivates discharge-time injection of phenotype tokens (Section ??) rather than relying on multi-admission temporal trends.

3 Method

3.1 Delphi backbone reproduction

Delphi-Base follows the original Delphi architecture: a 5.83 M-parameter, GPT-2-style causal Transformer with 12 layers, 12 heads, embedding dimension 192, block size 128, and 1,537 tokens covering ICD-10 diagnoses, CKD-stage tokens (CKD-EPI 2021 eGFR), age and sex tokens, and sentinels. Training optimises the joint cross-entropy and Weibull time-to-event loss ($\mathcal{L} = \mathcal{L}_{CE} + \mathcal{L}_{DT}$) with AdamW ($\beta_2 = 0.99$, weight decay 0.1, grad clip 1.0), batch size 128, and 14,500 total fine-tuning steps after vocabulary warm-up.

3.2 Discrete phenotype-token extension (Delphi-Px)

Delphi-Px adds 33 tokens covering four clinical domains: cancer status and treatment (14), psychiatric crisis (8), substance use (4), and high-acuity critical care (7). Tokens include `CANCER_STAGE_IV`, `SEPSIS_PRESENT`, `INTUBATED_DURING_STAY`, `COMFORT_MEASURES_ONLY`, `DNR_PRESENT`, and `ICU_ADMISSION`. They are extracted by a section-aware, negation-handling, rule-based pipeline that scans HPI, Hospital Course, Assessment, Plan, and Discharge Diagnoses sections, rejecting matches whose 80-character left context fires a negation cue (“no”, “denies”) or future-intent cue (“planned”, “recommend”). On 331,793 notes the extractor recovers 242,601 occurrences across 41.3% of notes in 188 seconds (30 CPU workers).

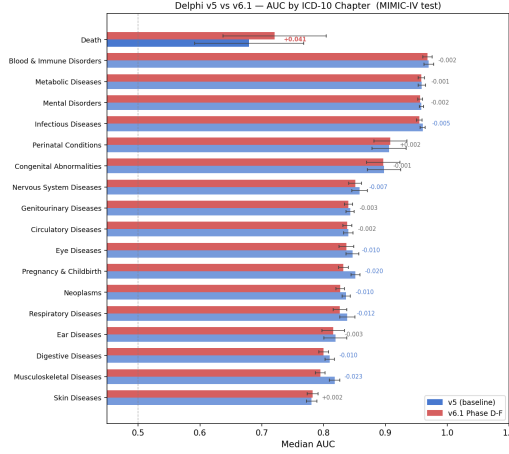
3.3 Leakage prevention and fine-tuning

Phenotype tokens are inserted into `train.bin`, `val.bin`, and `test_input.bin` but *not* into `test_future.bin`, so they appear as input context for test patients without ever serving as prediction labels. The vocabulary is expanded from 1,537 to 1,570; new rows are initialised near $\bar{w}_{ICD} + \epsilon$. Fine-tuning runs in two phases: **D-E** (1,000 steps, frozen backbone, only the new $33 \times 192 = 6,336$ embedding parameters trainable, LR 10^{-3}) and **D-F** (3,000 steps, all 5.84 M parameters trainable, LR 10^{-5}). A continuous alternative (*NoteProjector*) that mapped 768-D Clinical-Longformer embeddings into the 192-D Delphi space exhibited representation collapse (cosine similarity 0.9977; only 0.324 after diversity regularisation), no net ICD gain, and no token-level audit trail; we abandoned that branch in favour of the discrete approach.

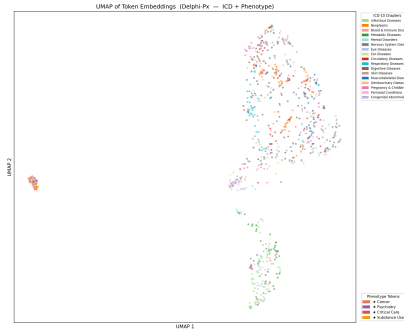
4 Experimental Results

4.1 Reproduction of the original Delphi figures

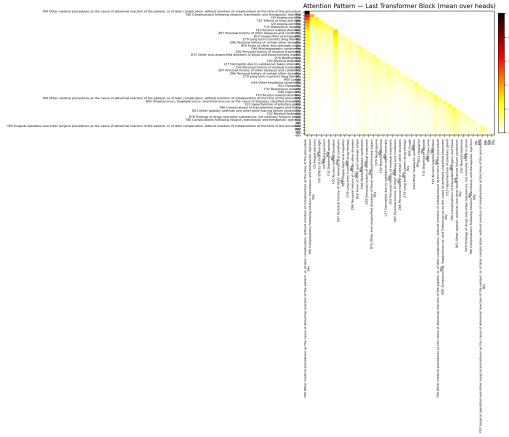
Figure ?? shows our four reproductions side by side. The per-chapter AUC panel (??) reproduces the chapter ordering from the original work, with Blood & Immune Disorders, Metabolic Diseases, and Mental Disorders highest and Skin and Musculoskeletal Diseases lowest. The UMAP panel (??) shows that semantically related ICD chapters cluster despite training only on co-occurrence, again matching the qualitative pattern of the original. The attention heatmap (??) confirms the causal triangular structure, with attention concentrated on recent and clinically related events. The risk-curve panel (??) reproduces the elevated I20 (Angina) risk for diabetic versus control patients across the 30–85 age range.



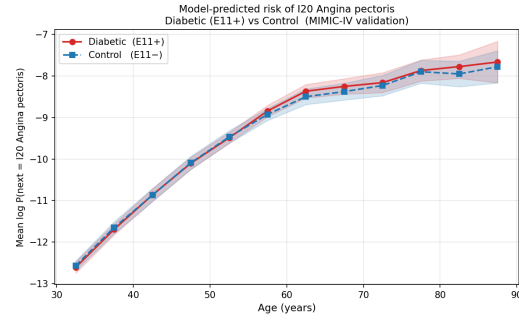
(a) Reproduction of Fig. 2f: per-chapter AUC.



(b) Reproduction of Fig. 3b: UMAP of 863 disease-token embeddings, coloured by ICD-10 chapter.



(c) Reproduction of Fig. 3c: causal attention heatmap from the last block, averaged over 12 heads, on a single validation patient with ≥ 40 events.



(d) Reproduction of Fig. 3d: predicted risk curves of I20 (Angina) for diabetic vs. control validation cohorts ($n = 300$ each), 5-year age windows with 95% CI.

Figure 2: Reproduction of figures 2f, 3b, 3c, 3d from Shmatko *et al.* (2025) using our MIMIC-IV-trained Delphi-Base. Chapter clustering, attention causality, and the diabetic–control separation all replicate qualitatively under the smaller and sparser MIMIC-IV cohort.

4.2 Benchmarking against the original paper

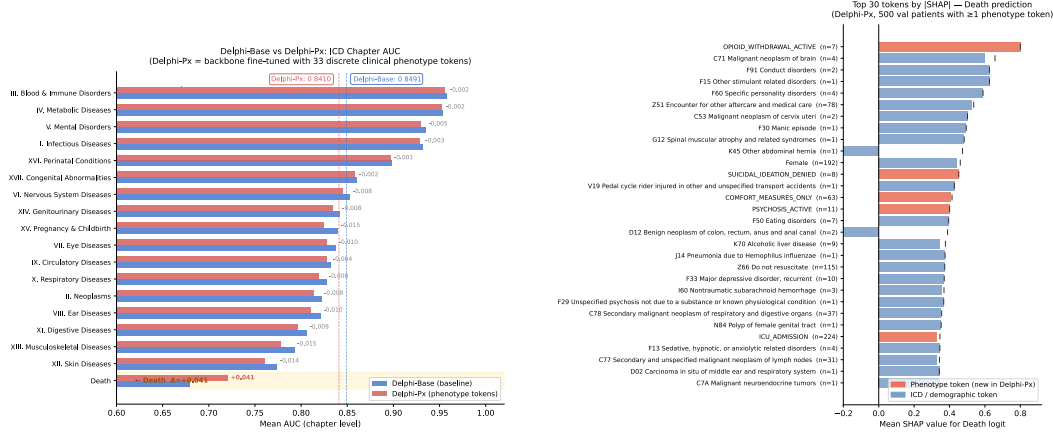
Table ?? compares our reproduction with the original paper. Delphi-Base achieves a higher mean chapter AUC (0.858) than the UK Biobank-trained Delphi (0.803), reflecting both the post-2015 ICD-10 cleanliness of MIMIC-IV and the narrower disease distribution of an ICU referral cohort. The Death AUC for our reproduction (0.680) is lower than the original paper’s UK Biobank Death AUC (0.730), consistent with the absence of multi-decade longitudinal follow-up in MIMIC-IV. Delphi-Px closes most of this gap (0.721) using only structured-event modeling plus 33 discrete note-derived tokens. Static baselines (LR, HGBT) on the same features achieve only 0.558–0.561, isolating the gain to sequence modeling rather than to feature augmentation. The C-index improves from 0.494 (Delphi-Base) to 0.533 (Delphi-Px), and 30-day mortality AUC from 0.486 to 0.525.

4.3 Per-chapter and SHAP results

Figure ?? shows that the 17 ICD chapters lose at most 0.016 each, with the largest decreases in Pregnancy & Childbirth (−0.016), Musculoskeletal (−0.015), and Skin (−0.014). This task-specific specialization is reflected in the SHAP attribution (Figure 2). All 33 phenotype tokens have

Table 1: Benchmarking our MIMIC-IV Delphi-Base reproduction and Delphi-Px extension against the original Shmatko *et al.* (2025) UK Biobank benchmarks. Chapter AUC is the unweighted mean over 17 ICD-10 chapters at filter ≥ 50 .

Model	Cohort	Mean Chapter AUC	Death AUC
Delphi (Shmatko <i>et al.</i> , 2025)	UK Biobank	0.803	0.730
Delphi-Base (our reproduction)	MIMIC-IV	0.858	0.680
Delphi-Px (this work)	MIMIC-IV	0.851	0.721
Logistic regression	MIMIC-IV	—	0.558
HGBT (+ ICD bag-of-words)	MIMIC-IV	—	0.561



(a) Per-chapter AUC for Delphi-Base (blue) versus Delphi-Px (orange). Death is the only column with a positive change (+0.041); each ICD chapter loses at most 0.016.

(b) Top tokens by absolute mean SHAP for Death prediction. Phenotype tokens (orange) are over-represented at the top relative to their share of the vocabulary; all 33 phenotype tokens have positive mean SHAP for Death.

Figure 3: Effect of phenotype-token addition on chapter-level AUC and SHAP-attribution patterns.

positive mean SHAP for Death, with an overall mean of +0.189. The top phenotype contributors are COMFORT_MEASURES_ONLY (+0.408), ICU_ADMISSION (+0.330), INTUBATED_DURING_STAY (+0.328), ALCOHOL_WITHDRAWAL_ACTIVE (+0.307), and DNR_PRESENT (+0.256). Among the top-30 tokens by absolute SHAP, five are phenotype tokens, although phenotype tokens account for only 33 of 1,570 model-space entries.

5 Conclusion

We reproduced the Delphi generative EHR transformer on MIMIC-IV with mean chapter AUC of 0.858 and replicated the original paper’s diagnostic figures (2f, 3b, 3c, 3d). We then extended the backbone with 33 discrete clinical phenotype tokens extracted from discharge summaries, raising Death AUC by +0.041 over the structured-only baseline while preserving token-level interpretability. SHAP attribution traces the gain to clinically meaningful high-acuity states rather than to opaque dense note features. External validation, broader phenotype coverage, and calibrated survival modeling are natural next steps. Code and the full extraction pipeline are publicly available at <https://github.com/AllyssaDrew/Delphi-Px-EHR-Foundation-model-with-MIMIC-IV>.

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